CHEMICAL EQUILIBRIUM

1. $K_p = K_c (RT)^{\Delta ng}$
   $K_p = K_x (P_T)^{\Delta ng}$

* $\ln a = 2.303 \log a$

2. Le Chatlier → jo bhi change aaya ⇒ usko nullify karo

a) P ↑ ⇒ Eq. chahega P ↓ ⇒ uss taraf mudd jaega jahan (n_T) kamm hain.

b) V → usko pehle P ki question mein change kardo.

c) Conc^n → usi taraf (★ solids ki conc^n se farak xxx)

d) T ↑ → Eq. chahega T ↓ ⇒ uss taraf mudd jaega jahan heat khai jaegi
↳ endo

e) Inert gas →

const. V
⇓
xxx

const. P
⇓
milke Vol ↑ such that Pr. remains const.
⇓
milke P ↓ hone laga
⇓
Eq. chahega P ↑
⇐ uss taraf mudd jaega jahan (n_T) zyada

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3. Haber's Process → $N_2(\uparrow) + 3H_2(\uparrow) \rightleftharpoons 2NH_3(\uparrow)$

favourable at → P_{High} + T_{low} → 700K

200 atm

* Catalyst – $Al_2O_3 + K_2O +$ (Fe oxide)

earlier → Fe (MO as catalytic Promoter)

4. Arhenious Eqⁿ

$$k = A\, e^{-\Delta t/RT}$$

i.e. $$\ln k = \frac{Ea}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

5. Van't Hoff Eqⁿ

$$K_{eq} = A' e^{-\Delta H/RT}$$

i.e $$\boxed{\ln \frac{K_{eq1}}{K_{eq2}} = \frac{\Delta H}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)}$$

6. Vapour Density = $\frac{\text{Density of Gas}}{\text{Density of } H_2}$ { at same T and P } = $\frac{MM}{2}$

7. $dG = V\,dP - S\,dT$ → at const Temp → $G = G^\circ + nRT \ln P$

$\Delta G_{Real} = \Delta G^\circ_{Real} + RT \ln Q$ → at eq. → $\Delta G = 0 \Rightarrow \Delta G^\circ_R = -RT \ln K_p$

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ } → This eqⁿ assumes ΔS° is independent of T